

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Bachelor of Pharmacy (B. Pharm)

Semester – II University Theory Examination May 2016

PH108 Pharmaceutical Chemistry II (Physical)

Date: 05/05/2015, Thursday

Time: 10:00 a.m. To 01:00 p.m. Maximum Marks: 80

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book is to be used.
3. Draw neat sketches wherever necessary.

SECTION- I

- Q 1** Attempt any **TWO** of the following:
- A** Give detail note on application of various radioproducts in diagnosis and treatment of various diseases. **10**
- B** State and explain the law of photochemistry. What do you mean by quantum yield? How is it determined experimentally? **10**
- C** **Answer the following.**
- 1) Write a note on partition coefficient. **05**
- 2) Enlist Colligative properties. Explain elevation of Boiling point. **05**
- Q 2** Attempt any **TWO** of the following:
- A** What is mean by Phase rule? Write a brief note on One component three phase system in detail. **10**
- B** What is Jablonski diagram? Draw and explain the Jablonski diagram. **10**
- C** **Answer the following.**
- 1) What is viscosity co-efficient? Describe Ostwald's viscometer for determining it. **05**
- 2) Explain Debye-Huckel theory. **05**

SECTION- II

- Q 3** Attempt any **TWO** of the following:
- A** Define the following terms: **10**
- I) Surface Tension, II) Viscosity, III) Refractive Index and IV) Parachor and Optical Rotation
- B** **Write a brief note on:** **10**
- 1) Sodium as an important Electrolyte **05**
- 2) Joule-Thomson Effect **05**
- C** What is meant by the term catalyst? Give general characteristics of catalytic reactions and Explain in detail about Homogeneous and Heterogeneous Catalysis with Suitable Examples. **10**

- Q 4** Attempt any **TWO** of the following:
- A** **Answer the following.**
- 1) Magnesium as an important Electrolyte **05**
 - 2) Differentiate Ideal Solution and Real solution in detail. **05**
- B** Write note on Carnot cycle and derive equation of thermodynamic efficiency. **10**
- C** Define surface tension. What is its unit? Describe the methods for the determination of the surface tension. **10**
-